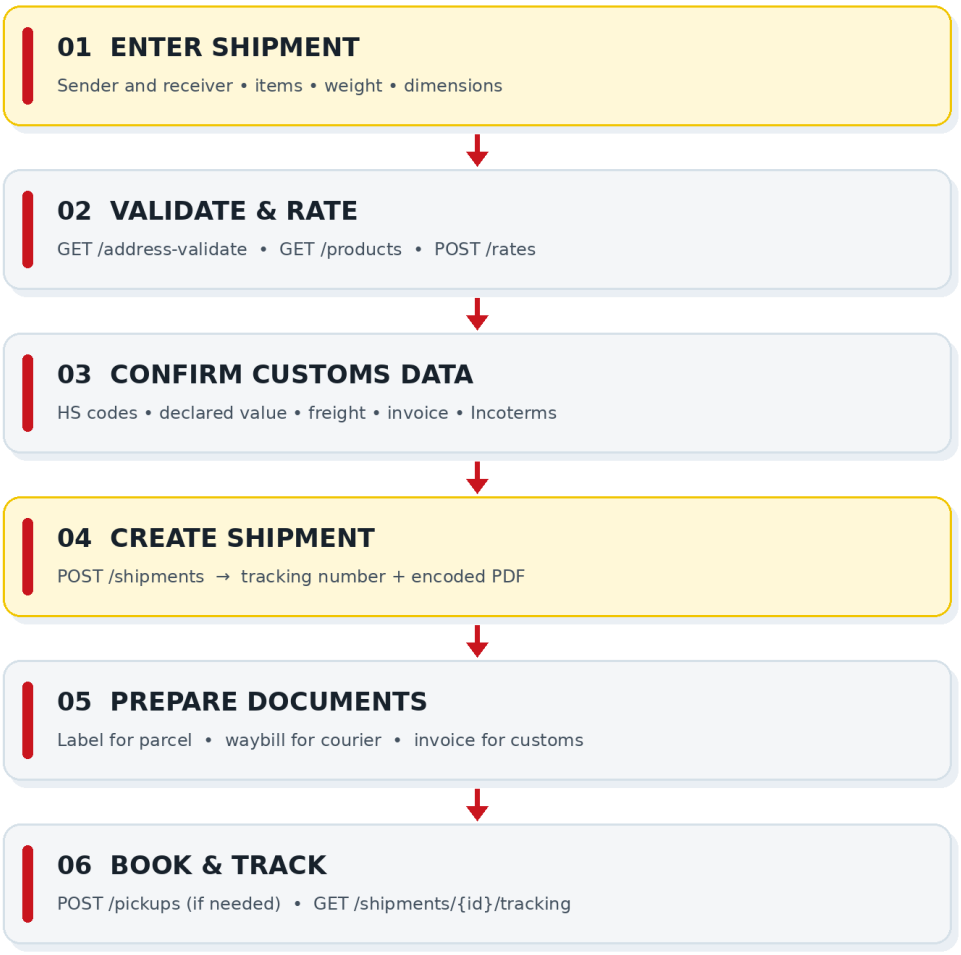
DHL / EXPRESS INTEGRATION

MyDHL API Integration Review & Action Plan

Requirements • Shipping documents • API workflow • Testing • Open issues

Prepared 9 November 2026 | Scope: Nigeria and Ghana shipment examples

3 sample documents	13 implementation tasks	5 priority verification areas



Illustrated recommended MyDHL transaction flow (summary).

Basis of review: Dhl.mp4; MYDHL API.json; MYDHL STAMP.pdf; official DHL Express MyDHL REST API and Nigerian customs documentation pages.

1. Executive summary

The main work is to integrate DHL Express shipping services into an application and generate the correct documents for each shipment. The supplied collection demonstrates address validation, rates, shipments, pickups, tracking and products. The sample PDF demonstrates a label, a courier waybill and a commercial invoice.

A complete implementation should validate addresses, obtain account-specific rates and product availability, create shipments, save the tracking number and generated documents, correctly prepare customs data, support pickup booking and tracking, and pass test-to-production verification.

Documented deliverables

- Validate pickup and delivery locations before quoting or booking.
- Retrieve available products and account rates for the correct origin, destination, pieces and shipment date.
- Create shipments using the appropriate domestic, export or import account and document/package configuration.
- Generate and distribute the correct label, waybill and commercial invoice; preserve the original DHL PDF output.
- Handle customs classification, declared value, freight, currency, insurance and optional electronic trade documents.
- Book pickup explicitly when required, store booking confirmation, and provide end-to-end tracking.
- Validate all examples, correct authentication, and complete integration testing before production.

Reference sources and interpretation

Reference	Evidence used
S1. MYDHL API.json	Local Postman-style collection with API requests, sample payload fields, product codes and service operations.
S2. MYDHL STAMP.pdf	Three-page DHL sample with label, waybillDoc, and commercial invoice for Nigeria → Ghana.
S3. Dhl.mp4	Visual demonstration of Postman requests, generated documents, success responses and a pickup credentials error.
D1. DHL Express MyDHL API	Official REST API operations, environment URLs, Basic Authentication and service descriptions.
D2–D3. DHL customs pages	Customs invoice data and Paperless Clearance context.

Evidence boundary: the video was reviewed visually, but meeting audio was not reliably transcribed. Any comments about what a presenter “said” should therefore be treated as unverified. Source-derived facts are distinguished from recommended implementation actions.

What success looks like

For one tested Nigeria-to-Ghana commercial parcel, DHL returns a successful shipment response and tracking number, the application stores the label and courier waybill, the commercial invoice reconciles to the contents and correct customs valuation, and pickup and tracking can be demonstrated.

2. The three DHL documents

2.1 Shipping label — page 1 of sample PDF

The parcel label identifies shipper and receiver, country routing, account/service indicators, piece count, weight, shipment reference and machine-readable DHL barcodes. The uploaded example is a sample and bears a “DO NOT PRINT - SAMPLE ONLY” overprint.

Required action: Generate a fresh label from a successful production shipment, preserve its barcode quality and attach the actual label to the parcel. Never use this demonstration copy for live shipping.

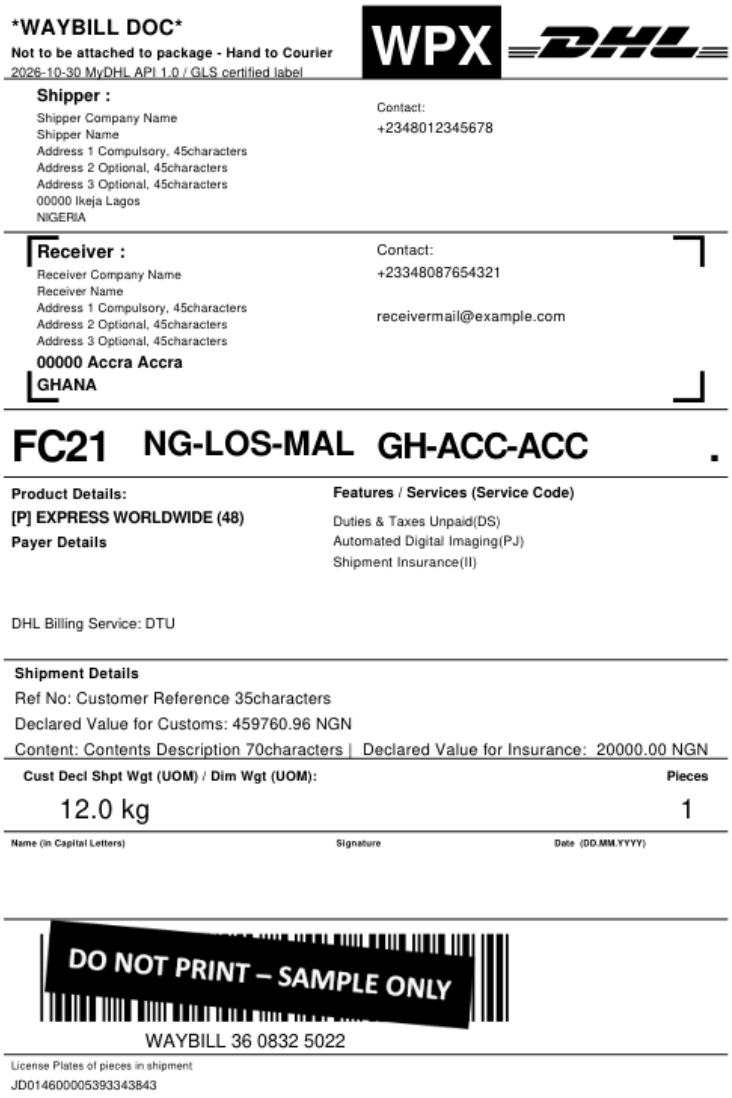


Figure 1. Actual page 1 of the uploaded MYDHL STAMP.pdf (sample DHL shipping label).

2.2 Courier waybill document — page 2 of sample PDF

This document contains the sender and receiver contact details, route, product/services, billing information, shipment weight and piece count, references, declared values and waybill identifiers.

Required action: Make this document available for download/printing and operational handover. The sample explicitly states: “Not to be attached to package - Hand to Courier.” The specific handover procedure should be confirmed with DHL.



WAYBILL DOC

Not to be attached to package - Hand to Courier

2026-10-30 MyDHL API 1.0 / GLS certified label

WPX

Shipper :

Shipper Company Name
Shipper Name
Address 1 Compulsory, 45characters
Address 2 Optional, 45characters
Address 3 Optional, 45characters
00000 Ikeja Lagos
NIGERIA

Contact:
+2348012345678

Receiver :

Receiver Company Name
Receiver Name
Address 1 Compulsory, 45characters
Address 2 Optional, 45characters
Address 3 Optional, 45characters
00000 Accra Accra
GHANA

Contact:
+23348087654321

receivermail@example.com

FC21 NG-LOS-MAL GH-ACC-ACC

Product Details:
[P] EXPRESS WORLDWIDE (48)
Payer Details

Features / Services (Service Code)
Duties & Taxes Unpaid(DS)
Automated Digital Imaging(PJ)
Shipment Insurance(II)

DHL Billing Service: DTU

Shipment Details
Ref No: Customer Reference 35characters
Declared Value for Customs: 459760.96 NGN
Content: Contents Description 70characters | Declared Value for Insurance: 20000.00 NGN

Cust Decl Shpt Wgt (UOM) / Dim Wgt (UOM): 12.0 kg

Pieces: 1

Name (in Capital Letters) Signature Date (DD.MM.YYYY)

DO NOT PRINT – SAMPLE ONLY

WAYBILL 36 0832 5022

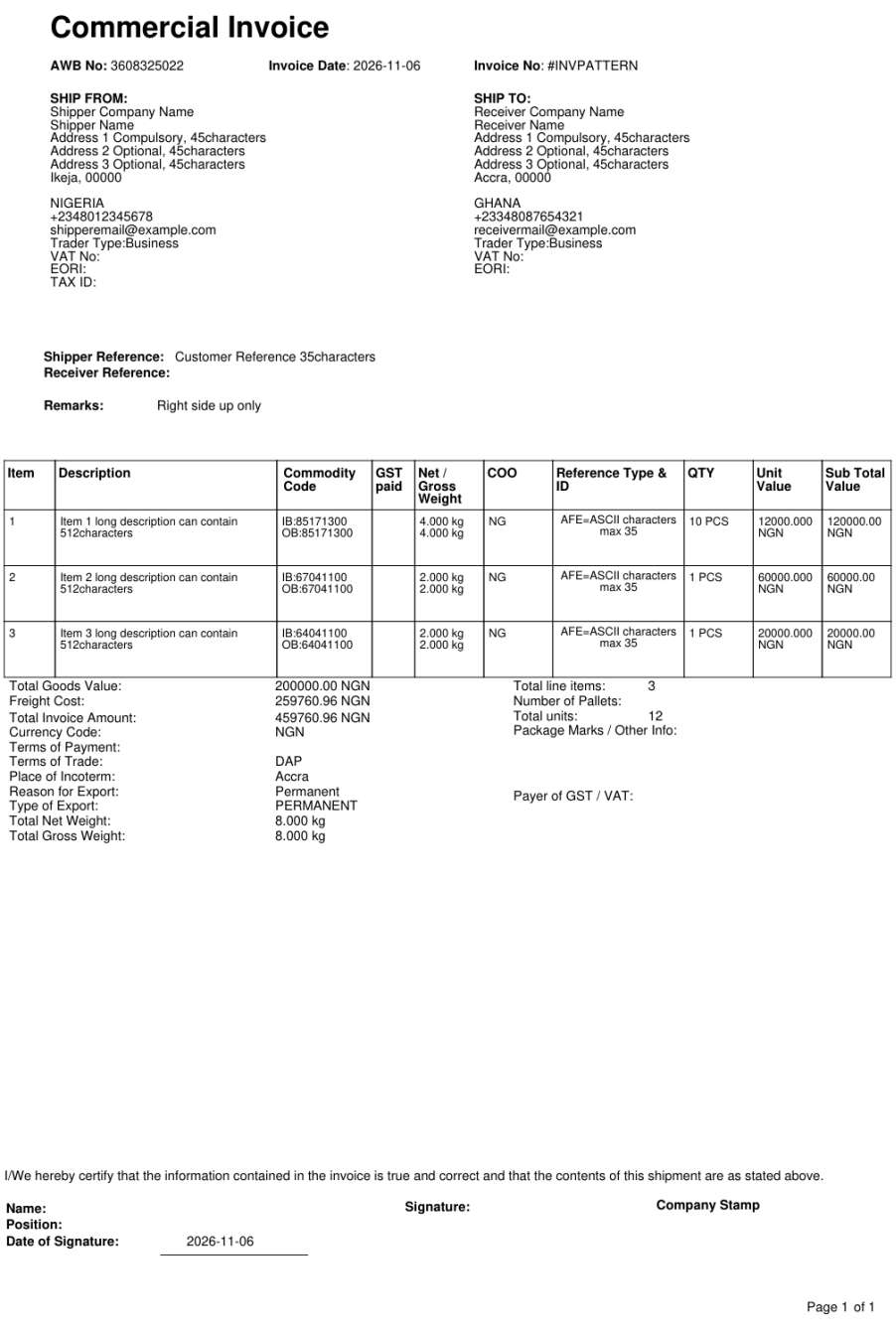
License Plates of pieces in shipment
JD014600005393343843

Figure 2. Actual page 2 of the uploaded MYDHL STAMP.pdf (courier waybill document).

2.3 Commercial invoice — page 3 of sample PDF

The customs invoice identifies the shipper and receiver and lists the goods in itemized form. The sample includes item descriptions, commodity codes, country of origin, quantities, values, freight, invoice total, payment/trade terms, the reason for export and the declaration sign-off area.

Required action: Ensure goods, weights, codes, values and charges match the actual shipment. Provide a signature and/or stamp when required for the country, DHL workflow and shipment type. Do not assume all shipments require a physical stamp: eligibility for electronic customs documents must be confirmed.



Commercial Invoice

AWB No: 3608325022 **Invoice Date**: 2026-11-06 **Invoice No**: #INVPATTERN

SHIP FROM:
Shipper Company Name
Shipper Name
Address 1 Compulsory, 45characters
Address 2 Optional, 45characters
Address 3 Optional, 45characters
Ikeja, 00000

NIGERIA
+2348012345678
shipperemail@example.com
Trader Type:Business
VAT No:
EORI:
TAX ID:

SHIP TO:
Receiver Company Name
Receiver Name
Address 1 Compulsory, 45characters
Address 2 Optional, 45characters
Address 3 Optional, 45characters
Accra, 00000

GHANA
+23348087654321
receivermail@example.com
Trader Type:Business
VAT No:
EORI:

Shipper Reference: Customer Reference 35characters
Receiver Reference:

Remarks: Right side up only

Item	Description	Commodity Code	GST paid	Net / Gross Weight	COO	Reference Type & ID	QTY	Unit Value	Sub Total Value
1	Item 1 long description can contain 512characters	IB:85171300 OB:85171300		4.000 kg 4.000 kg	NG	AFE=ASCII characters max 35	10 PCS	12000.000 NGN	120000.00 NGN
2	Item 2 long description can contain 512characters	IB:67041100 OB:67041100		2.000 kg 2.000 kg	NG	AFE=ASCII characters max 35	1 PCS	60000.000 NGN	60000.00 NGN
3	Item 3 long description can contain 512characters	IB:64041100 OB:64041100		2.000 kg 2.000 kg	NG	AFE=ASCII characters max 35	1 PCS	20000.000 NGN	20000.00 NGN

Total Goods Value: 200000.00 NGN
Freight Cost: 259760.96 NGN
Total Invoice Amount: 459760.96 NGN
Currency Code: NGN
Terms of Payment:
Terms of Trade: DAP
Place of Incoterm: Accra
Reason for Export: Permanent
Type of Export: PERMANENT
Total Net Weight: 8.000 kg
Total Gross Weight: 8.000 kg

Total line items: 3
Number of Pallets:
Total units: 12
Package Marks / Other Info:

Payer of GST / VAT:

I/We hereby certify that the information contained in the invoice is true and correct and that the contents of this shipment are as stated above.

Name:
Position:
Date of Signature: 2026-11-06

Signature:

Company Stamp

Page 1 of 1

Figure 3. Actual page 3 of MYDHL STAMP.pdf. The signature/company-stamp areas are visible but not completed in the sample.

3. API mapping and PDF generation

The supplied collection uses POST /shipments for the shipment transaction and requests documents through outputImageProperties. DHL returns tracking information and encoded document content; the implementation should decode and store the original PDFs, index them by shipment, and expose safe download/print controls.

Collection setting	Role / implementation rule
encodingFormat: "pdf"	Request document output in PDF format.
allDocumentsInOneImage: true	Request combined document output; still identify each generated document accurately.
typeCode: "label"	DHL shipping label for placement on parcel.
typeCode: "waybillDoc"	Courier waybill document; in sample, hand to courier.
typeCode: "invoice"	Commercial invoice / customs paperwork when appropriate.
hideAccountNumber: true	Hide the account number in the waybill output as configured.
pickup.isRequested: false	No pickup is requested as part of shipment creation in these examples.

Requested sample template identifiers

- ECOM26_84_A4_001 — label
- ARCH_8X4_A4_002 — waybillDoc
- COMMERCIAL_INVOICE_P_10 — invoice

These are template choices in the uploaded collection, not a universal requirement for every DHL account. Check supported formats, account approval and output behavior with DHL.

API endpoints at a glance

Operation	Request	Purpose
Address validation	GET /address-validate	Pickup/delivery capability by location.
Products	GET /products	Available DHL services for eligible shipments.
Rates	POST /rates	Account rating for multi-piece shipments.
Shipment	POST /shipments	Create shipment and return labels/documents.
Pickup	POST /pickups	Book courier pickup (when required).
Tracking	GET /shipments/{id}/tracking	Retrieve tracking events and status.

DHL's official service reference also documents other operations (including pickup changes/cancellations and customs invoice upload). The collection covers a practical subset of that wider interface. [D1]

4. Implementation backlog

The checklist below translates the source examples and public API requirements into developer work packages. P0 is the minimum integration, P1 completes documentation/operations, and P2 is acceptance and go-live readiness.

P0 — Core integration

☐ **Configure credentials and environments —** Securely configure Basic Authentication, test/production URLs and approved import/export accounts.

☐ **Validate addresses —** Implement GET /address-validate with clear address validation feedback.

☐ **Retrieve available products and rates —** Retrieve product capabilities and POST /rates using actual dimensions, weights, service date, and accounts.

☐ **Create shipments —** Implement POST /shipments for eligible domestic, export, and import scenarios; handle document/package rules.

☐ **Store outputs —** Save tracking and piece numbers, decode returned PDF content, and provide download/print actions.

P1 — Customs and operations

☐ **Commercial invoice mapping —** Build itemized invoice data: goods, HS codes, quantity, price, origin, freight, terms and values.

☐ **Invoice sign-off —** Support review, approval and signature/company stamp when operationally required.

☐ **Paperless Trade assessment —** Verify eligibility and electronic upload/submission route with DHL.

☐ **Pickup booking —** POST /pickups, store confirmation and implement appropriate change/cancellation actions.

☐ **Tracking —** Retrieve shipment tracking details and display status and exceptions.

P2 — QA and launch

☐ **Payload and security validation —** Remove placeholders and JSON comments, correct auth overrides, enforce HTTPS and field validation.

☐ **End-to-end test matrix —** Test all supported routes, weights/pieces, customs decisions, output documents and negative responses.

☐ **DHL production readiness —** Obtain DHL account and format approvals and complete the agreed technical acceptance process.

5. Issues identified and remediation

Issue 1 — Pickup authentication error

The video frame at approximately 14:04 shows POST /pickups returning HTTP 400 and the message “Invalid Credentials”. The local collection declares noauth on seven pickup requests and on two shipment requests. The collection-level auth object is empty, so credential inheritance must be explicitly configured in the executing tool. [S1, S3]

Action: Set up pre-emptive Basic Authentication according to DHL’s REST guide and use consistent request-level inheritance; verify account permissions and environment/credential pairing. Re-run the pickup test before diagnosing any other fields. [D1]

Issue 2 — Shipment and invoice weight mismatch

The uploaded label/waybill shows shipment weight as 12.0 kg, while the invoice totals show 8.000 kg net and 8.000 kg gross. Packaging weight could explain some difference, but the relationship is not stated in the sample. [S2, pp. 1–3]

Action: Capture product net weight, item gross weight and fully packed shipment weight separately. Define and test consistent rules for each field, including any packaging weight.

Issue 3 — Declared value vs. invoice total

The collection’s export package example sets goods declaredValue to NGN 200,000 and freight charges to NGN 259,760.96; the invoice reports a total of NGN 459,760.96. The sample courier waybill describes NGN 459,760.96 as “Declared Value for Customs”. [S1, S2]

Action: Separate goods value, freight, total invoice amount, insurance and customs value. Do not automatically equate invoice total and customs value; ask DHL to confirm the correct valuation basis for this route.

Issue 4 — Example request hygiene

The supplied bodies contain sample placeholders, inline // comments in many JSON examples, hard-coded reference values and a Postman strictSSL setting of false. These are demonstration fixtures, not production-ready values. [S1]

Action: Emit valid JSON without comments; replace every placeholder with validated inputs; use correct current dates, genuine commodity codes and per-shipment references; preserve TLS certificate checks and redact credentials from logs.

Issue 5 — Customs and pickup decision rules

Some document examples request an invoice despite isCustomsDeclarable being false. The shipment examples also specify pickup.isRequested=false. Domestic, export/import and document/goods cases need separate eligibility rules. [S1]

Action: Define when customs paperwork is necessary; when physical vs. electronic invoice submission is supported; and when to book a pickup through POST /pickups. DHL confirmation is required for exceptional situations. [D1–D3]

6. Visual evidence from the supplied video

The stills below are cropped from Dhl.mp4 to focus on the shared application and documents rather than meeting participants. They are visual observations only, not a transcript of the spoken session.

6.1 Shipment request and successful response

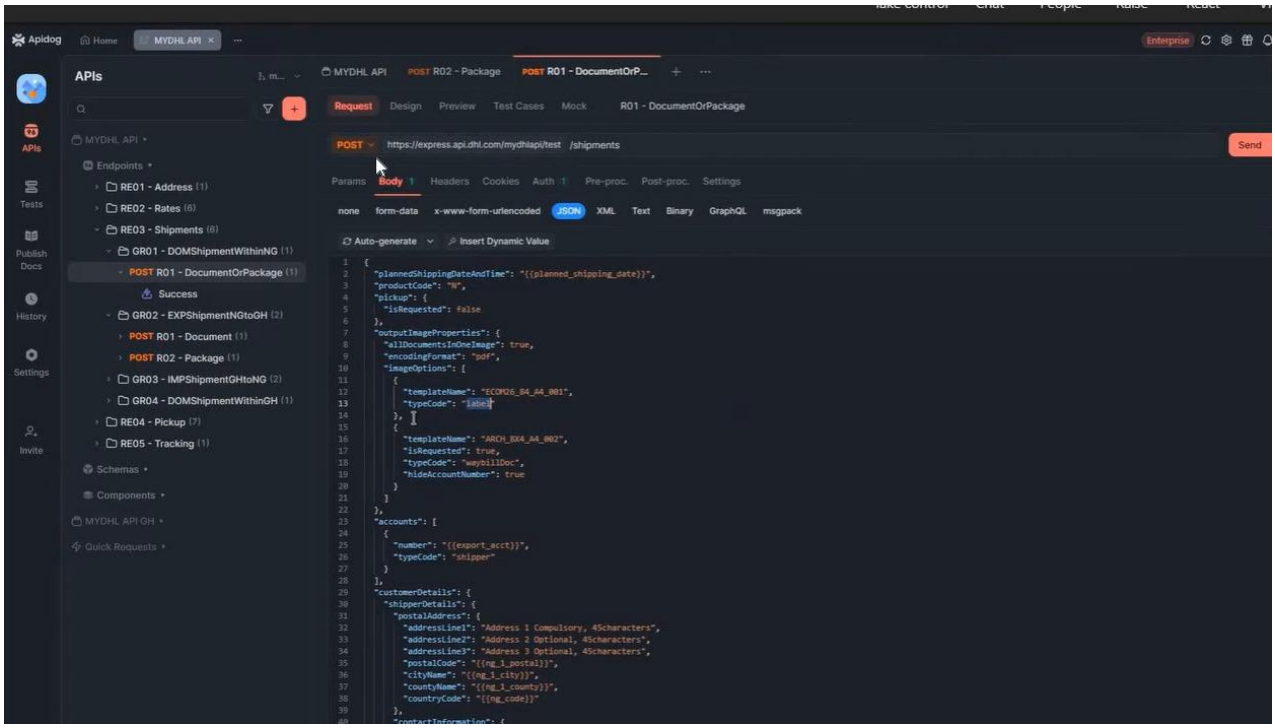


Figure 4. Around 00:08: POST /shipments request, including label and waybill output configuration.

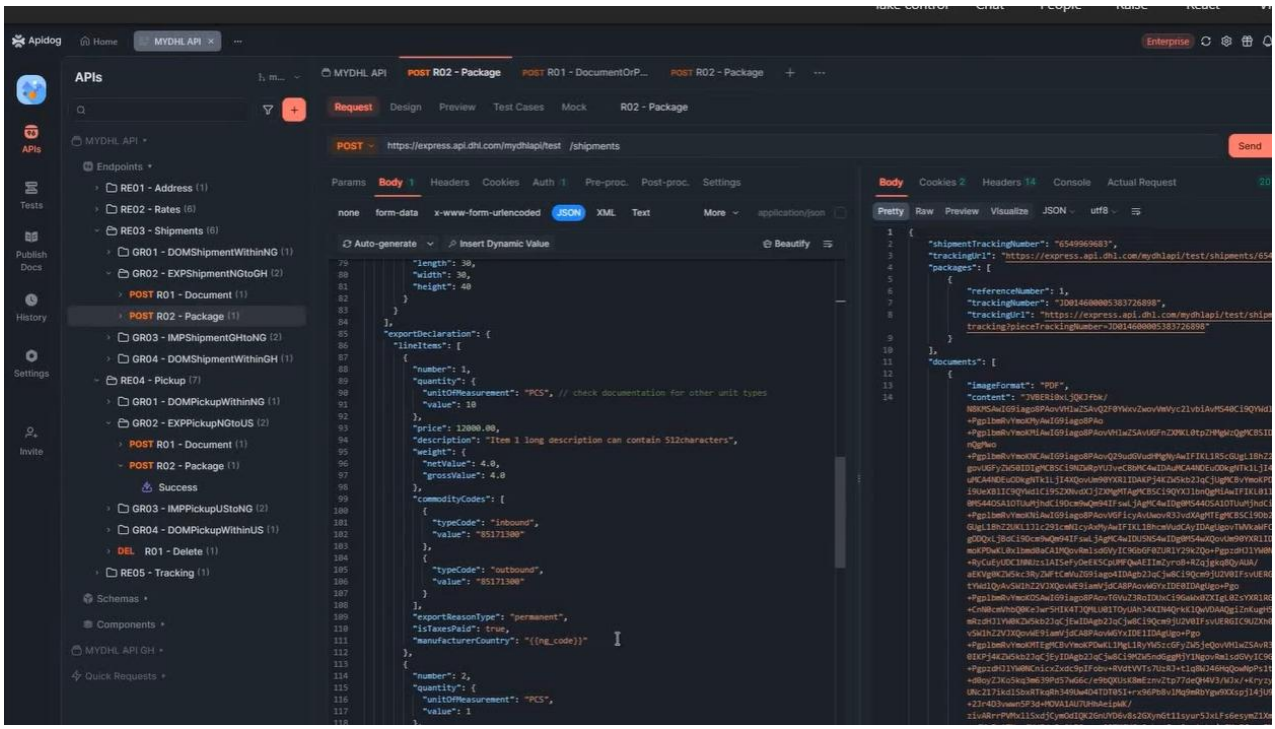


Figure 5. Around 10:30: POST /shipments response with HTTP 201, shipmentTrackingNumber and encoded PDF data.

6.2 Authorization and pickup error

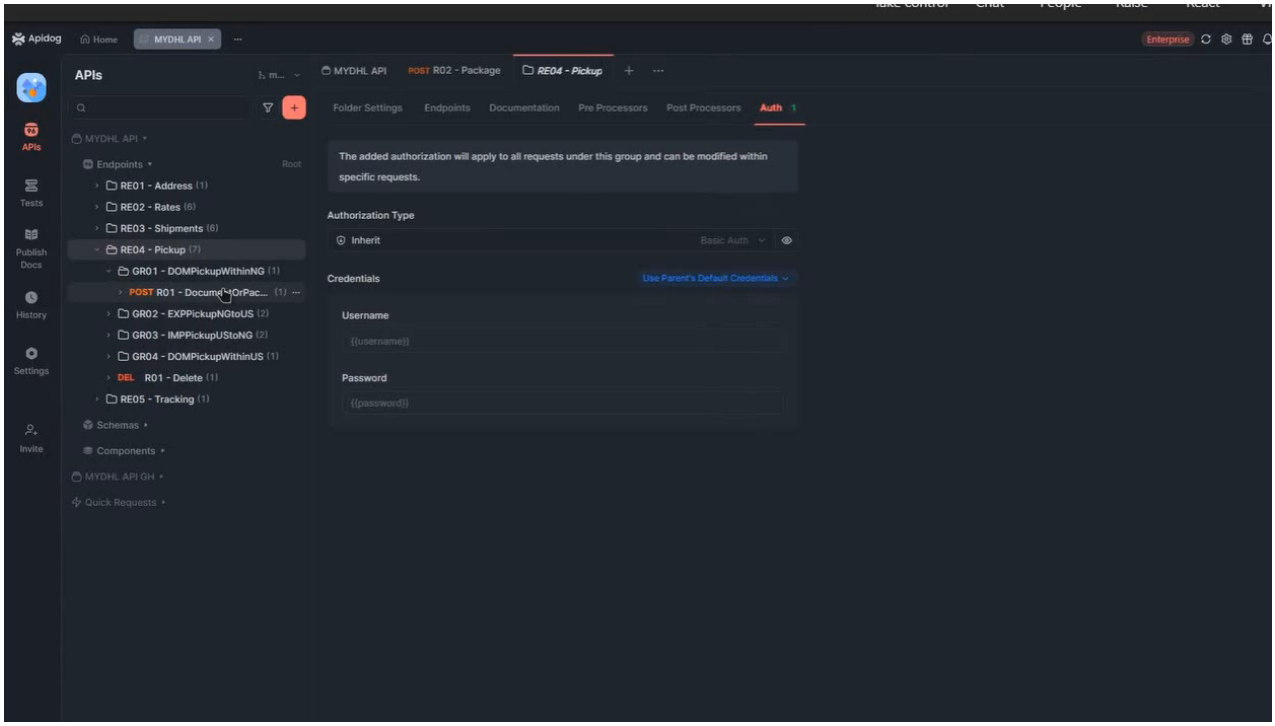


Figure 6. Around 13:35: group authorization is shown as inherited Basic Auth credentials.

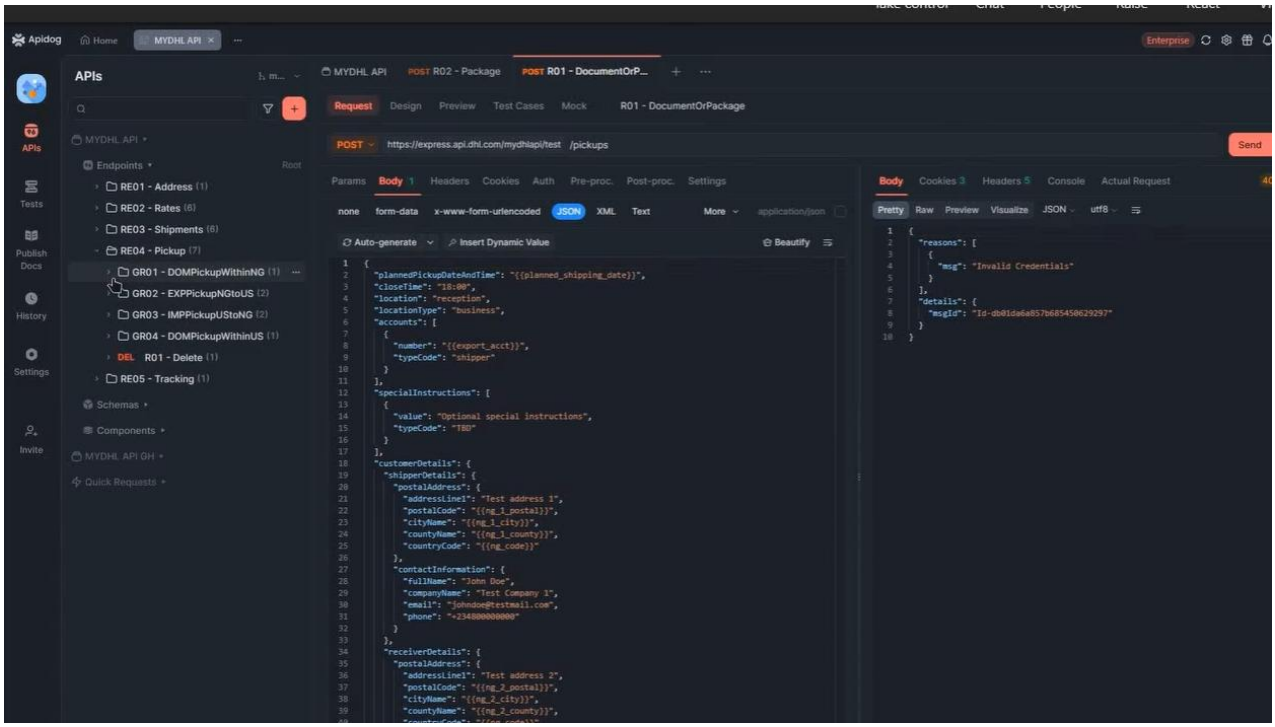


Figure 7. Around 14:04: pickup test returns HTTP 400 with “Invalid Credentials”.

6.3 Documents demonstrated in the video

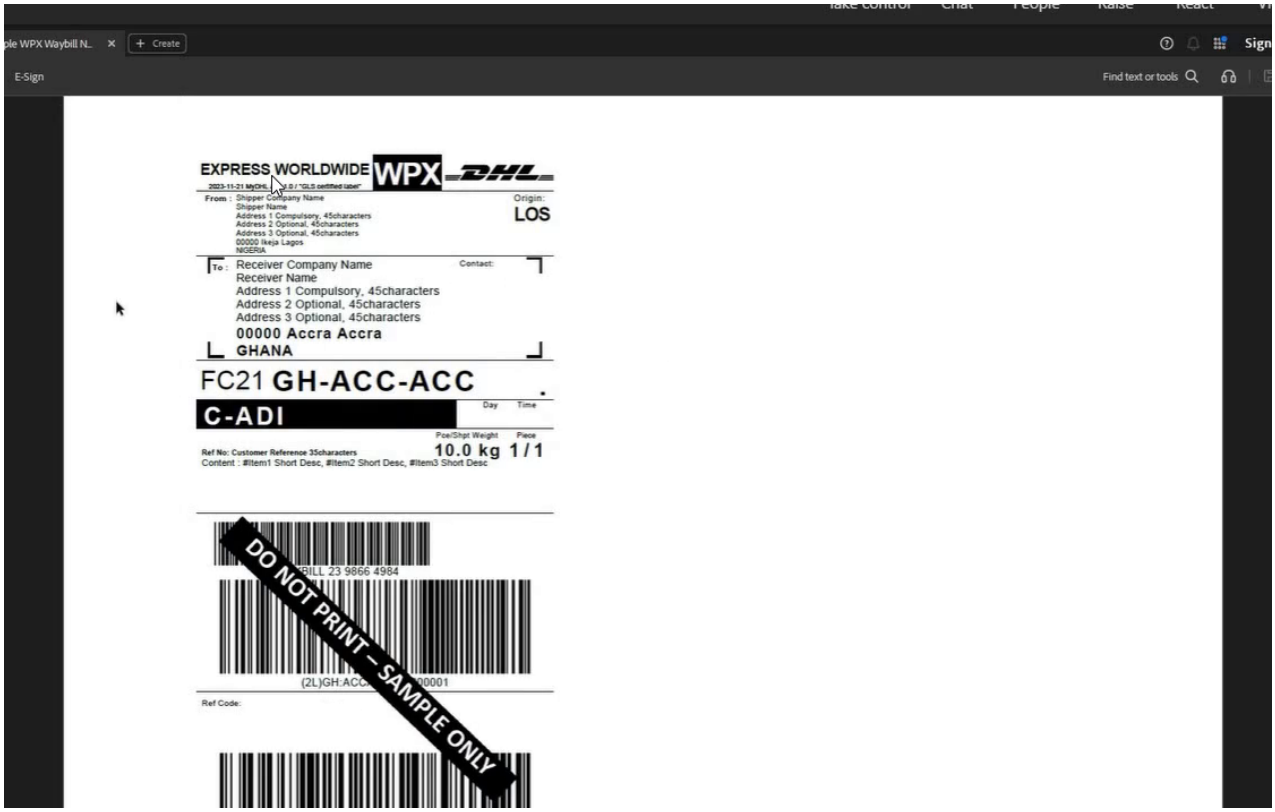


Figure 8. Around 02:52: generated sample DHL shipping label is opened in a PDF viewer.

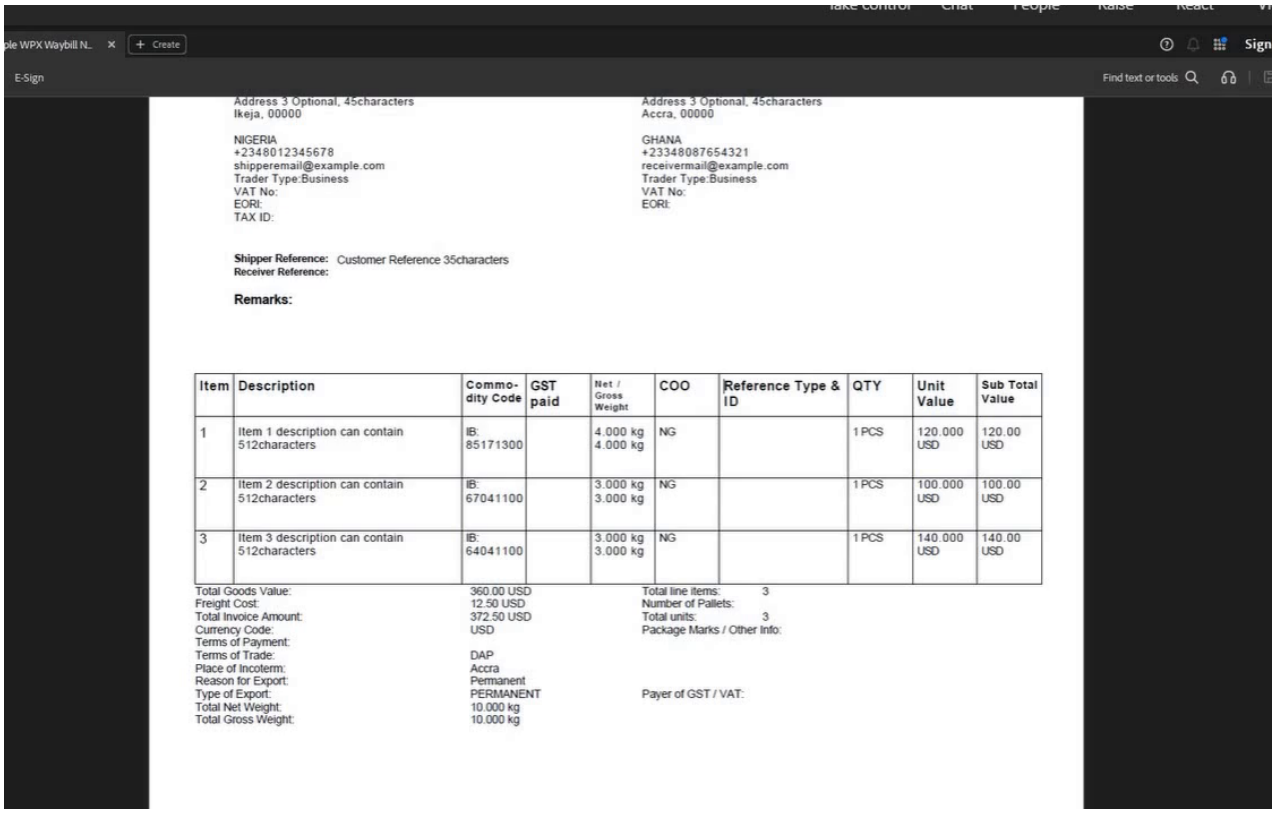


Figure 9. Around 08:55: a commercial invoice is demonstrated in a PDF viewer. This screen uses another example and need not match the uploaded sample PDF values.

7. Recommended operating workflow

Use one consistent business flow across UI, backend API integration and documents. Only create a shipment after address, product/rate and customs information are validated. A shipping booking must be distinguished from actual handover to DHL.

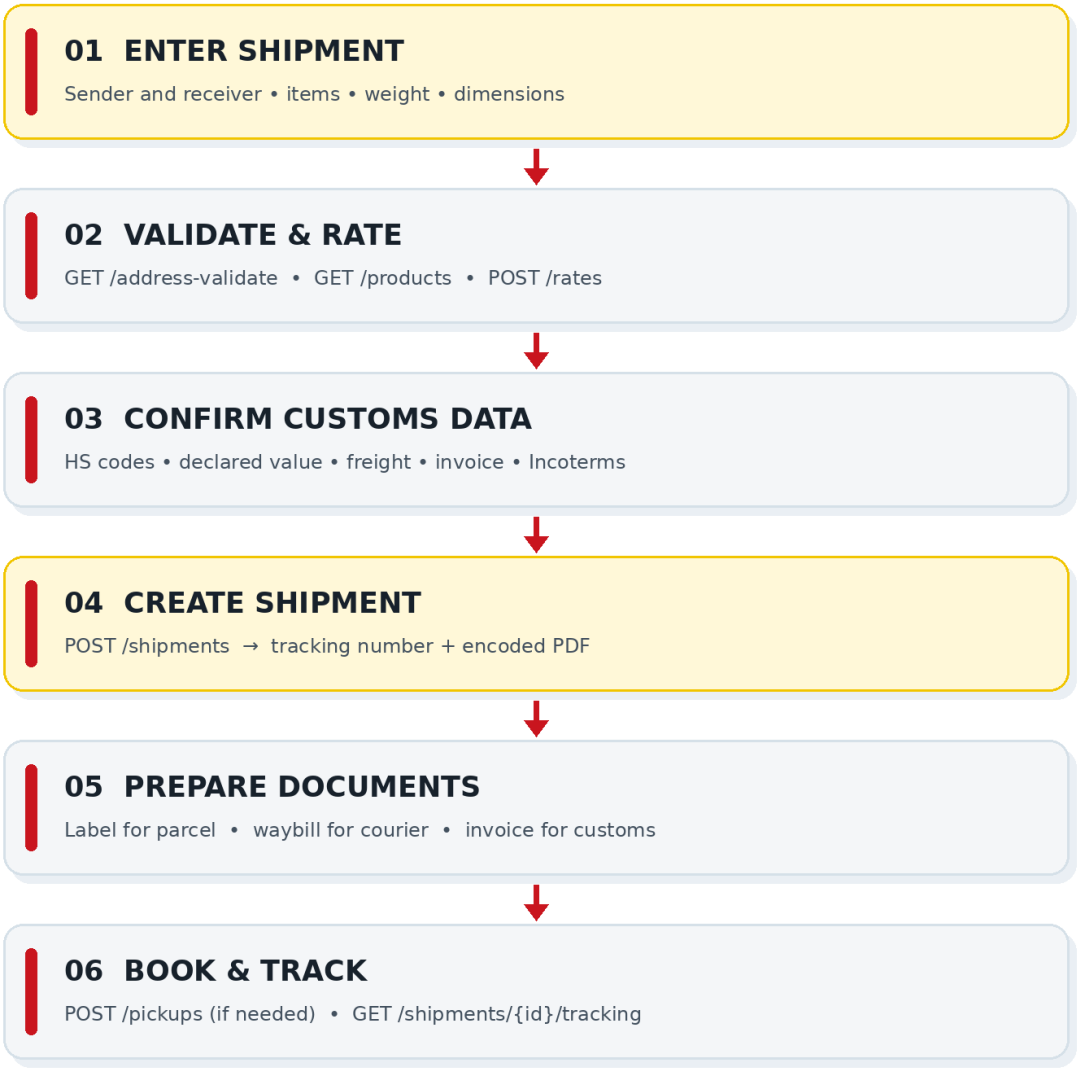


Figure 10. Implementable shipment process derived from the integration requirements.

Operational distinction: pickup.isRequested=false in the shared shipment requests means that shipment creation is not itself a pickup booking in those examples. Confirm the explicit pickup step and operational handover procedure. [S1, D1]

8. End-to-end test plan and acceptance criteria

Scenario	Expected result / evidence
Domestic delivery in Nigeria	Correct location validation, rate, domestic product, tracking, parcel label and waybill.
Nigeria → Ghana: document	Correct supported document product; necessary transport and customs documents.
Nigeria → Ghana: commercial parcel	Correct goods/HS codes, consignee, currencies, customs invoice, weight and freight reconciliation.
Ghana → Nigeria: import	Correct import account, payer-country configuration and customs information.
Multi-piece shipment	Piece counts, total weight, labels and piece tracking identifiers correspond to the package list.
Pickup request	Authenticated booking succeeds; confirmation number is recorded; errors are handled.
Tracking	Track using a real generated reference, update statuses and display exceptions.
Document QA	Label barcode readable; courier waybill instructions followed; invoice signed/stamped when required.
Negative cases	Bad credentials, invalid address, unsupported product, malformed JSON and inconsistent data fail clearly.

Acceptance evidence to retain

- ☐ Sanitized request/response examples for every supported shipment route.
- ☐ Saved DHL tracking number and all generated PDF documents from each representative case.
- ☐ Reconciled weights, invoice line items, freight, goods values and currency.
- ☐ Proof that authentication errors and schema errors are displayed and logged safely.
- ☐ DHL sign-off on account configuration, customs handling, label template and production readiness.

9. Questions requiring DHL confirmation

- ☐ When are physically signed/stamped invoices necessary for the intended origin/destination pairs?
- ☐ Is Paperless Trade enabled for the DHL account and eligible for these commodity and route combinations?
- ☐ Are ECOM26_84_A4_001, ARCH_8X4_A4_002 and COMMERCIAL_INVOICE_P_10 the intended production templates?
- ☐ Which export/import account, billing payer type and value-added services should be used for each scenario?
- ☐ How should the NGN 200,000 goods value and NGN 259,760.96 freight be reported for customs valuation?

10. Sources and official references

User-provided evidence

S1. MYDHL API.json — uploaded API request collection (Address, Rates, Shipments, Pickup, Tracking, Products).

S2. MYDHL STAMP.pdf — uploaded three-page DHL sample label, courier waybill and commercial invoice.

S3. Dhl.mp4 — uploaded recorded walkthrough. Figures 4–9 are still frames from the video.

Public DHL documentation

D1. MyDHL API (DHL Express), REST API reference — https://developer.dhl.com/api-reference/mydhl-api-dhl-express?language_content_entity=en

D2. DHL Nigeria: customs clearance documents — https://www.dhl.com/ng-en/home/express/shipping-and-tracking/customs/customs-clearance/customs-clearance-documents.html

D3. DHL Nigeria: paperless clearance — https://www.dhl.com/discover/en-ng/ship-with-dhl/export-with-dhl/paperless-clearance

D4. DHL Nigeria: customs clearance FAQ — https://www.dhl.com/ng-en/home/express/shipping-and-tracking/customs/customs-clearance.html

Proposed completion sequence

1. Fix credentials and establish a working test account. 2. Validate address and rates. 3. Complete one commercial parcel shipment with correct customs data. 4. Validate all three document outputs. 5. Add pickups/tracking. 6. Run the full test matrix and secure DHL approval.

Important: this document is an implementation review and not an official DHL certification or customs ruling. Account-specific service eligibility, documentary rules, and live charges must be verified with DHL before production use.